CSE202 OOP MID-TERM COOKBOOK

150 practice questions • 50 per unit • C++

Concept MCQs | *Code output* | *Missing code* | *Debugging* | *Application*

Instructions for this practice set

- This is a revision practice bank based on the uploaded CSE202 syllabus for Units I, II and III. It is not an official university paper or a guarantee of exact exam questions.
- Each question has four options (A–D). Select one best answer. Treat each question as 1 practice mark; total = 150 practice marks.
- Attempt one unit at a time in 30–35 minutes. Do not look at the answer key until you finish that unit.
- For code-output questions, assume standard C++ with the usual headers and using namespace std unless a header is shown. Whitespace is significant only when explicitly described.
- For missing-code questions, choose the option that makes the snippet or stated intention correct. For debugging questions, choose the best diagnosis or correction.
- Revision order for tonight: Unit I functions/classes, Unit II pointers/arrays/strings, Unit III constructors/file modes/file pointers. Re-attempt every wrong answer.

Syllabus coverage

- Unit I: OOP basics, cin/cout, classes and objects, struct/union/enum, inline and static members, procedural vs OOP, default arguments, manipulators, overloading, scope, friends, references, parameter passing and recursion.
- Unit II: void and other pointers, pointer arithmetic, pointer-to-pointer, dangling/wild/null pointers, pointers in classes, object pointers, this pointer, arrays and array of objects, multidimensional arrays, string class and pointer-to-data-member.
- Unit III: file streams and modes, open/close, reading/writing, sequential/random and binary file processing, class/struct file operations, constructors, destructors, copy constructor and initializer lists.

UNIT I — Concepts and Basics of C++ Programming; Functions

50 questions • Suggested time: 30–35 minutes • Answer key at the end

Core concepts and syntax

1. Which statement best describes object-oriented programming?

A) A program organized only around functions
B) A program organized around objects that combine data and behavior
C) A program that cannot use functions
D) A program that works only with databases

2. Which keyword begins a class definition in C++?

A) object
B) define
C) class
D) structclass

3. By default, members of a C++ class are:

A) public
B) protected
C) private
D) global

4. By default, members of a C++ structure are:

A) private
B) public
C) protected
D) static

5. Which operator is used with cin to extract input?

A) <<
B) >>
C) ::
D) &&

6. Which operator is used with cout to insert output?

A) >>
B) <<
C) ->
D) %

7. Which OOP feature bundles data and the functions that operate on it?

A) Encapsulation
B) Compilation
C) Recursion
D) Iteration

8. Which paradigm commonly emphasizes procedures/functions operating on data?

A) Object-oriented
B) Procedural
C) Event-only
D) Declarative-only

9. Which type stores only one of its non-static data members in the same memory at a time?

A) class
B) struct
C) union
D) enum

10. What does an enum primarily define?

A) A group of named integral constants
B) A dynamic array
C) A file stream
D) A member function

Code output

11. What is printed?

```
int add(int a=2, int b=3) { return a+b; }
int main() { cout << add(); }
```

A) 5
B) 6
C) 23
D) Compilation error

12. What is the output?

```
int f(int x) { return x+1; }
double f(double x) { return x+0.5; }
int main() { cout << f(3) << " " << f(3.0); }
```

A) 4 3.5
B) 3 3.0
C) 4 4
D) Compilation error

13. What is printed?

```
inline int square(int x) { return x*x; }
int main() { cout << square(4); }
```

A) 8
B) 16
C) 4
D) Compilation error

14. What is the output?

```
void change(int x) { x=9; }
int main() { int x=4; change(x); cout << x; }
```

A) 9
B) 4
C) 5
D) Compilation error

15. What is printed?

```
void change(int &x) { x=7; }
int main() { int n=4; change(n); cout << n; }
```

A) 4
B) 7
C) 0
D) Compilation error

16. What is printed by the recursive function?

```
int fact(int n) { if(n<=1) return 1; return n*fact(n-1); }
int main() { cout << fact(4); }
```

A) 6
B) 24
C) 12
D) Infinite recursion

17. What is printed?

```
class A { public: static int x; };
int A::x=5;
int main() { A a; cout << a.x; }
```

A) 0
B) 5
C) Garbage value
D) Compilation error

18. What is the visible output (spaces shown as dots)?

```
int main() { cout << setw(4) << 7 << "X"; }
```

A) ...7X
B) 7...X
C) 7X
D) 7X

19. What is printed?

```
class A { int x=20; friend void show(A); };
void show(A a) { cout << a.x; }
int main() { A a; show(a); }
```

A) 10
B) 20
C) 0
D) Compilation error

20. What is printed?

```
int mul(int a, int b=4) { return a*b; }
int main() { cout << mul(3); }
```

A) 7
B) 12
C) 3
D) Compilation error

Missing code

21. Fill the blank to declare a class named Box.

A) struct Box()
B) class Box { };
C) Box class;
D) define class Box;

22. Complete the output statement for displaying variable n using cout.

A) cout >> n;
B) cout << n;
C) cin << n;
D) print(n);

23. Which line correctly declares a function with a default value for its second parameter?

A) int f(int a=1, int b);
B) int f(int a, int b=2);
C) int f(int a=1, int b=2, int c);
D) int f(int a, =2);

24. Fill the blank so calling bump(x) can modify the caller variable.

A) void bump(int x)
B) void bump(int *x) but use x=9
C) void bump(int &x)
D) void bump(const int &x)

25. Which declaration marks a member function as inline within the class definition?

A) inline: void show();
B) void inline show();
C) void show() { cout << x; }
D) friend show();

26. Fill the blank to make get() a static member function.

A) virtual int get();
B) static int get();
C) friend static get();
D) int static: get();

27. Which declaration correctly specifies a friend function inside class A?

A) friend void display(A);
B) friend class display(A);
C) void friend(A) display;
D) friend A::display();

28. Choose the correct call to invoke a function named show with no parameters.

A) show;
B) call show();
C) show();
D) void show;

29. Fill the blank to pass x by address to change(int *p).

A) change(x);
B) change(&x);
C) change(*x);
D) change(ref x);

30. Choose the correct stream manipulator that inserts a newline and flushes the stream.

A) setw
B) setprecision
C) endl
D) ws

Debugging and behavior

31. Which overload is selected by the call f(2)?

```
void f(int) { cout << "int"; }
void f(double) { cout << "double"; }
int main() { f(2); }
```

A) The double overload, because double is larger
B) Both overloads are called
C) The int overload, because the argument is an exact match
D) Neither overload can accept an int

32. Which declaration is invalid because of default-argument placement?

A) int f(int a, int b=3);
B) int f(int a=2, int b=3);
C) int f(int a=2, int b);
D) int f(int a, int b);

33. What is a likely issue with this code?

```
int f(int n) { return n*f(n); }
```

A) It is a correct recursive factorial
B) It never reaches a base case and recurses indefinitely
C) It returns 1 for every input
D) Recursion cannot be used in C++

34. Which statement about a friend function is correct?

A) It must be a member of the class
B) It is not a member, but may access private/protected members
C) It can access only public members
D) It cannot accept class objects

35. Which is true about a reference variable?

A) It can be left uninitialized
B) It must be initialized when declared
C) It can be reseated to refer to another object
D) It is always a pointer object stored separately

36. Which parameter form permits a function to modify the original integer directly?

A) int x
B) const int &x
C) int &x
D) int x[] only

37. Which statement about inline functions is most accurate?

A) The compiler must always substitute the body at every call
B) inline is a request/permission, not a guarantee of inlining
C) An inline function cannot have parameters
D) An inline function cannot be defined in a class

38. Which scope resolution operator is used to define a class member function outside its class?

A) .
B) ->
C) ::
D) *

39. What is required to use setw(5) in standard C++?

A) #include <vector>
B) #include <iomanip>
C) #include <string>
D) #include <fstream>

40. A function that calls itself is using:

A) Overloading
B) Encapsulation
C) Recursion
D) Manipulation

Applied concepts and higher-yield checks

41. Which is a valid benefit of using classes?

A) They prevent all runtime errors
B) They group data and related operations and support access control
C) They eliminate the need for constructors in every program
D) They make all members public automatically

42. Which statement about a union is true?

A) All data members have separate storage
B) All members share the same storage region
C) A union cannot contain numeric members
D) Union members are always private

43. A named set of constants such as RED, GREEN, BLUE is naturally represented by:

A) enum
B) friend
C) reference
D) union only

44. What is the primary role of a constructor?

A) Destroy an object
B) Initialize an object when it is created
C) Print an object automatically
D) Allocate a file name only

45. What is the usual effect of making data members private?

A) They cannot be accessed directly from unrelated outside code
B) They cannot be accessed by member functions
C) They become static
D) They are stored in a separate file

46. Which call passes a variable by value?

A) f(x) where f is declared f(int x)
B) f(&x) where f takes int*
C) f(x) where f takes int&
D) f(std::ref(x)) only

47. Which call passes an address to a pointer parameter int* p?

A) f(x)
B) f(*x)
C) f(&x)
D) f(int x)

48. Which statement about static data members is true?

A) Each object has its own separate copy
B) The member is shared by all objects of the class
C) It can only be accessed from constructors
D) It must be declared with auto

49. Which stream is normally used for standard error messages?

A) cin
B) cout
C) cerr
D) clogin

50. Which is the best contrast between procedural and object-oriented design?

A) Procedural design centers on functions; OOP centers on objects/classes
B) OOP cannot use functions
C) Procedural programs cannot have data
D) Both mean exactly the same thing

UNIT II — Pointers, Reference Variables, Arrays and String Concepts

50 questions • Suggested time: 30–35 minutes • Answer key at the end

Core concepts and syntax

1. What does a pointer variable store?

A) A function result only
B) A memory address
C) Always a whole array
D) Only a character

2. Which pointer type can hold the address of an object of any data type, but must be converted before dereferencing?

A) void*
B) nullptr_t* only
C) string*
D) reference*

3. What does nullptr represent in modern C++?

A) A null pointer literal convertible to any pointer type
B) An uninitialized pointer
C) A dangling pointer
D) A reference with no object

4. What is a pointer-to-pointer declared as for an int pointer p?

A) int &p
B) int **pp
C) int *&pp only
D) int pp[]

5. If int a[5];, what does a typically convert to in an expression?

A) int value of first element
B) pointer to its first element
C) pointer to last element
D) reference to whole array in all contexts

6. Which operation advances an int pointer p to the next int element?

A) p++
B) p=p+sizeof(int)*sizeof(int)
C) ++(*p)
D) p--

7. A pointer that refers to an object whose lifetime has ended is:

A) Null pointer
B) Dangling pointer
C) Void pointer
D) Reference pointer

8. A pointer that has not been initialized and contains an indeterminate address is commonly called:

A) Wild pointer
B) Smart pointer
C) Void pointer
D) Array pointer

9. Which operator accesses a member through a pointer to an object?

A) .
B) ::
C) ->
D) &

10. Inside a non-static member function, this points to:

A) The current object
B) The parent class only
C) The next object
D) The current function code

Code output

11. What is printed?

```
int a[]={10,20,30}; int *p=a;
int main() { cout << *p; }
```

A) 10
B) 20
C) 30
D) Undefined behavior

12. What is printed?

```
int a[]={10,20,30}; int *p=a;
int main() { cout << *(p+1); }
```

A) 10
B) 20
C) 30
D) Compilation error

13. What is printed?

```
int a[]={5,10,15}; int *p=a;
int main() { p+=2; cout << *p; }
```

A) 5
B) 10
C) 15
D) Compilation error

14. What is printed?

```
int x=7; int *p=&x; int **pp=&p;
int main() { ++(**pp); cout << x; }
```

A) 7
B) 8
C) 9
D) Compilation error

15. What is printed?

```
int x=3; int &r=x;
int main() { r++; cout << x << " " << r; }
```

A) 3 4
B) 4 3
C) 3 3
D) Compilation error

16. What is printed?

```
string s="Cat";
int main() { s="Dog"; cout << s; }
```

A) Cat
B) Dog
C) CatDog
D) Compilation error

17. What is printed?

```
string s="code";
int main() { cout << s.size(); }
```

A) 4
B) 5
C) 6
D) Compilation error

18. What is printed?

```
#include <cctype>
string s="code";
int main() { s[0]=toupper(s[0]); cout << s; }
```

A) code
B) Code
C) ode
D) Compilation error

19. What is printed?

```
int a[2][2]={{1,2},{3,4}};
int main() { cout << a[1][0]; }
```

A) 2
B) 3
C) 4
D) Compilation error

20. What is printed?

```
class A { public: int x=8; };
int main() { A a; A *p=&a; cout << p->x; }
```

A) 8
B) 9
C) 10
D) Compilation error

Missing code

21. Fill the blank to declare a pointer to int.

A) int p*;
B) int *p;
C) pointer int p;
D) int &*p;

22. Fill the blank to make p a null pointer.

A) int *p = 0xABC;
B) int *p = nullptr;
C) int *p = unassigned;
D) int p = nullptr;

23. Given int x=5; choose a pointer assignment that stores x's address.

A) int *p=x;
B) int *p=&x;
C) int &p=&x;
D) int p=&x;

24. Given int *p and int x, which expression writes 9 into x through p when p=&x?

A) p=9;
B) *p=9;
C) &p=9;
D) **p=9;

25. Fill the blank to access member value through A* p.

A) p.value
B) p::value
C) p->value
D) p&value

26. Choose the correct declaration for a two-row, three-column int array.

A) int a[2,3];
B) int a[2][3];
C) int a(2)(3);
D) array<int,2,3> a;

27. Fill the blank to append text t to string s.

A) s.add(t);
B) s.append(t);
C) s.push(t);
D) s.insertAll(t);

28. Fill the blank to find the first occurrence of "ab" in string s.

A) s.find("ab")
B) s.search("ab")
C) s.locate("ab")
D) find(s,"ab")

29. Which declaration makes a pointer-to-pointer to int?

A) int *p;
B) int **pp;
C) int &*r;
D) int &&pp;

30. Which syntax declares a pointer to a data member named x of class A, where x is int?

A) int A::x*;
B) int A::* pm;
C) A::*int pm;
D) int *A.x;

Debugging and behavior

31. What is the issue with dereferencing p here?

```
int *p=nullptr;
cout << *p;
```

A) No issue
B) p is nullptr, so dereferencing it has undefined behavior
C) nullptr automatically becomes 5
D) The code creates a reference

32. Which action can create a dangling pointer?

A) Assigning nullptr to a pointer
B) Keeping a pointer to a local variable after the function returns
C) Incrementing a valid pointer within an array
D) Using a pointer immediately after initialization

33. Which is the safest correction for a pointer after deleting its dynamic allocation?

A) Increment it
B) Set it to nullptr
C) Dereference it once more
D) Convert it to void*

34. Which statement about a void* is correct in standard C++?

A) It may be directly dereferenced without conversion
B) It can store an object address, but cannot be directly dereferenced as a typed object
C) It is identical to int*
D) It always points to nullptr

35. Given int a[4], which pointer comparison is within the same array bounds for normal element access?

A) a+4 dereferenced
B) a+3 dereferenced
C) a-1 dereferenced
D) a+100 dereferenced

36. Which statement about references and pointers is correct?

A) A reference may be reseated freely
B) A pointer must always refer to a valid object
C) A reference must be initialized and cannot be reseated
D) A pointer cannot be null

37. Which statement about a pointer to an object is correct?

A) Use -> to access a member through the pointer

B) Use :: for all members

C) Use * to access every member directly

D) Use # to access members

38. What does s.substr(1, 3) return for string s="abcdef"?

A) "abc"

B) "bcd"

C) "cde"

D) "bcde"

39. What does s.erase(2, 2) do to s="abcdef"?

A) Removes the first two characters

B) Removes two characters beginning at index 2, yielding "abef"

C) Removes index 2 only

D) Erases the whole string

40. What is the principal risk of this assignment?

```
class A { public: int *p; };
int *q = new int(5); A a; a.p=q; A b=a;
```

A) It is always safe

B) The class may copy the pointer value rather than the pointed-to resource, causing ownership problems

C) Pointers cannot be class members

D) It automatically duplicates the pointed-to object

Applied concepts and higher-yield checks

41. For string s="hello", what does s[1] refer to?

A) The first character h

B) The second character e

C) The last character o

D) The length 5

42. Which string member function returns the number of characters?

A) length() or size()

B) capacityOnly()

C) countChars()

D) strlen() as a member

43. Which string operation replaces a substring at a specified position?

A) replace()

B) rewind()

C) clearbit()

D) truncateFile()

44. Which declaration creates an array of three objects of class Student?

A) Student s(3);

B) Student s[3];

C) class Student[3] s;

D) array Student s;

45. If int a[2][3] is declared, valid second-dimension indices are:

A) 1 through 3
B) 0 through 2
C) 0 through 3
D) -1 through 1

46. If p points to a[0] in an int array, what does p+2 point to?

A) Two bytes later
B) a[2], assuming the pointer remains within the array
C) The address of the pointer variable p
D) Always a null pointer

47. Which is a correct way to declare a reference to an int?

A) int &r = x;
B) int r& = x;
C) ref int r=x;
D) int *&r=x;

48. What does the this pointer allow a member function to refer to?

A) The current object, often to distinguish a member from a parameter
B) A global object only
C) The base address of every class in the program
D) The next statement in memory

49. Which operation is often used to test whether a string contains a substring?

A) find() and compare against string::npos
B) fopen()
C) sizeof(string) only
D) pointer arithmetic on characters only

50. A pointer-to-data-member is best described as:

A) A pointer that stores a member value itself
B) A special pointer representing a data member within a class, applied to an object to access that member
C) A reference to a complete class definition
D) A function pointer only

UNIT III — Data File Operations; Constructors, Destructors and File Handling

50 questions • Suggested time: 30–35 minutes • Answer key at the end

Core concepts and syntax

1. Which header provides ifstream, ofstream, and fstream?

A) <iostream>
B) <fstream>
C) <iomanip>
D) <cstring>

2. Which class is commonly used to read from a file?

A) ofstream
B) ifstream
C) ostream
D) istream only

3. Which class is commonly used to write to a file?

A) ifstream
B) ofstream
C) istream
D) ios_base only

4. Which file mode opens a file for output and normally truncates an existing file?

A) ios::in
B) ios::out
C) ios::binary only
D) ios::ate only

5. Which mode positions the file pointer at the end when opening?

A) ios::beg
B) ios::ate
C) ios::trunc only
D) ios::noreplace

6. Which mode is used for binary file operations?

A) ios::binary
B) ios::textmode
C) ios::byte
D) ios::raw

7. Which function closes a file stream explicitly?

A) finish()
B) endfile()
C) close()
D) stop()

8. A constructor is called primarily to:

A) Initialize an object as it is created
B) Delete an object
C) Close every file automatically
D) Return an integer from a class

9. A destructor for class A is declared as:

A) ~A()
B) !A()
C) delete A()
D) A~()

10. A copy constructor typically has which parameter form?

A) A(A other, A other2)
B) A(const A& other)
C) void A(A*)
D) A(int other) only

Code output

11. What is printed?

```
class A { public: int x; A(int n):x(n){} };
int main() { A a(2); cout << a.x; }
```

A) 1
B) 2
C) 0
D) Compilation error

12. What is printed?

```
class A { public: int x; A():x(7){} };
int main() { A a; cout << a.x; }
```

A) 7
B) 0
C) Garbage value
D) Compilation error

13. What is printed?

```
class A { public: int x; A(int n):x(n){} A(const A& o):x(o.x){} };
int main() { A a(10); A b(a); cout << b.x; }
```

A) 5
B) 10
C) 15
D) Compilation error

14. What is printed?

```
class A { public: ~A(){ cout << "D"; } };
int main() { A a; cout << "C"; }
```

A) C
B) D
C) CD
D) Compilation error

15. What is the output?

```
class A { public: A(){cout<<"A";} ~A(){cout<<"B";} };
int main() { A a; }
```

A) AB
B) BA
C) AAB
D) Compilation error

16. Assume file "demo.txt" opens and contains "Hi". What does this code print?

```
ifstream fin("demo.txt"); string s; fin >> s; cout << s;
```

A) Hi
B) demo.txt
C) Nothing
D) Compilation error

17. Which result is most appropriate for this code if the file opened successfully and contains 3 lines?

```
ifstream fin("data.txt"); string line;
while (getline(fin, line)) { cout << line << "\n"; }
```

A) It reads exactly one line and stops forever
B) The loop processes input lines until getline fails at end-of-file
C) It always processes 4 lines
D) It cannot read text files

18. What is printed?

```
class A { public: int x; A(int n):x(n){} A(const A& z):x(z.x+1){} };
int main(){ A a(1); A b(a); cout<<a.x<<" "<<b.x; }
```

A) 1 2
B) 2 1
C) 1 1
D) Compilation error

19. What is the key effect of calling fin.close() after reading?

A) It deletes the file
B) It closes the associated file stream
C) It rewinds the file to the beginning
D) It converts text to binary

20. What does this constructor initializer list set?

```
class A { int x; public: A():x(4){} };
```

A) x to 4 before the constructor body executes
B) x to 4 only after destruction
C) x to a file position
D) Nothing; initializer lists are comments

Missing code

21. Fill the blank to open a file for reading.

A) ofstream fin("a.txt");
B) ifstream fin("a.txt");
C) fstream fin("a.txt", ios::out);
D) openstream fin("a.txt");

22. Fill the blank to open a file for writing.

A) ofstream fout("a.txt");
B) ifstream fout("a.txt");
C) istream fout("a.txt");
D) cin.open("a.txt");

23. Complete the statement to close file stream fin.

A) fin.end();
B) fin.close();
C) close(fin);
D) fin.stop();

24. Fill the blank for a default constructor of class A.

A) A(int);
B) ~A();
C) A() { }
D) void A();

25. Fill the blank to declare a destructor in class A.

A) A();
B) ~A();
C) delete A();
D) destructor A();

26. Choose a correct initializer list for x to be initialized to 10.

A) A() { x(10); }
B) A() : x(10) { }
C) A() -> x=10 { }
D) A() => x(10) { }

27. Fill the blank for a copy constructor that binds to a source object without copying it first.

A) A(A other)
B) A(const A& other)
C) A(const A other&)
D) A(int & other)

28. Which condition is commonly used to read formatted values until extraction fails?

A) while (fin >> x)
B) while (fin.close())
C) if (fin == eof)
D) while (x = fin.close())

29. Fill the blank to open a binary output file.

A) ofstream f("x.bin", ios::out | ios::binary);
B) ifstream f("x.bin", ios::text);
C) ofstream f("x.bin", ios::read);
D) fstream f("x.bin", ios::null);

30. Choose the correct call to move an output file stream position to byte offset 0 from the beginning.

A) f.seekp(0, ios::beg);

B) f.seekg(0, ios::end);

C) f.tellg(0);

D) f.close(0);

Debugging and behavior

31. What is wrong with this code if A has no default constructor?

```
class A { public: A(int n){} };
int main() { A a; }
```

A) Nothing; every class has an automatic default constructor

B) A a; is invalid because no zero-argument constructor is available

C) A class cannot declare constructors

D) The destructor is missing

32. Why should a copy constructor commonly accept const A& rather than A by value?

A) Passing by value could require the copy constructor again and cause recursive copying

B) References are not allowed for constructors

C) const makes all members static

D) It forces file output

33. Which statement about a destructor is true?

A) It can have parameters

B) It has the same name as the class prefixed by ~ and no return type

C) It must return int

D) Every destructor must be static

34. If an object is constructed successfully as a local variable inside a block, when is its destructor normally called?

A) Before its constructor

B) When the object leaves its scope

C) Only when main ends, regardless of scope

D) Never

35. Which file read pattern is generally safer than checking eof() before reading?

A) while (!fin.eof()) { fin >> x; }

B) while (fin >> x) { /* process x */ }

C) if (fin.eof()) fin >> x;

D) while (fin.close())

36. Why must binary file reads/writes of raw object memory be used carefully?

A) Every class is automatically portable as raw bytes

B) Objects with pointers, virtual functions, or non-trivial ownership may not be safely serialized as raw bytes

C) Binary mode only supports integers

D) Binary files cannot be opened with fstream

37. Which statement about ios::app is correct?

A) Every write is directed to the end of the file
B) It opens only for reading
C) It automatically deletes the file
D) It is identical to ios::in

38. Which file-position function is associated with the input/get pointer?

A) seekp
B) tellp
C) seekg
D) putback only

39. Which file-position function is associated with the output/put pointer?

A) seekg
B) tellg
C) seekp
D) getline

40. What is the typical issue with this statement if the class member is a const int?

```
class A { const int x; public: A() { x=5; } };
```

A) No issue; it is the recommended initializer syntax
B) A const data member should be initialized in the constructor initializer list, not assigned later in the body
C) const data members must always be global
D) The object will be deleted

Applied concepts and higher-yield checks

41. Which constructor is invoked by A b(a); when a is an existing object of class A?

A) Default constructor
B) Copy constructor
C) Destructor
D) File constructor

42. Which constructor is often invoked by A a(5); if A has A(int)?

A) Default constructor
B) Parameterized constructor
C) Copy constructor
D) Destructor

43. A constructor with default arguments can often serve as:

A) Only a destructor
B) Both a no-argument constructor call and calls with supplied arguments
C) A file stream only
D) A global variable declaration

44. Which object type supports both input and output file operations?

A) ifstream
B) ofstream
C) fstream
D) ostringstream only

45. Which access pattern is sequential file processing?

A) Always jumping directly to a chosen byte offset
B) Processing records in order from the current position
C) Accessing only the final record
D) Using only binary files

46. Which feature supports direct movement to a specified file position?

A) Random access using seekg/seekp
B) Function overloading
C) Friend class
D) Inline function

47. Why might a class that owns a dynamically allocated resource need a custom copy constructor?

A) To control how the resource is copied and avoid shallow-copy ownership bugs
B) To prevent every assignment operator from compiling by default
C) To turn the destructor into a static member
D) To permit cout

48. Which statement is true about an initializer list?

A) It initializes bases and data members before the constructor body
B) It runs after the destructor
C) It is used only with files
D) It cannot initialize const members

49. Which file mode commonly preserves existing contents while adding new output at the end?

A) ios::app
B) ios::trunc
C) ios::in only
D) ios::beg

50. When two local objects are constructed in the order A then B in the same scope, their destructors normally run in which order?

A) A then B
B) B then A
C) Random order
D) Neither destructor runs

ANSWER KEY + QUICK CHECKS

Check this only after attempting all questions in a unit. Explanations are brief; review the relevant concept if you missed one.

Unit 1 — Concepts and Basics of C++ Programming; Functions

1. B — Objects combine state (data) and behavior (functions).

2. C

3. C

4. B

5. B

6. B

7. A

8. B

9. C — Union members share storage.

10. A

11. A — The default arguments are 2 and 3.

12. A — The int overload returns 3+1; the double overload returns 3.0+0.5.

13. B — square(4) returns 4*4.

14. B — The function receives a copy, so x in main is unchanged.

15. B — The reference parameter aliases the original variable.

16. B — factorial(4) is 4*3*2*1 = 24.

17. B — The static member is defined and initialized to 5.

18. A — setw(4) sets the width of the next output field only.

19. B — The friend function is allowed to access the private member x.

20. B — The second argument uses its default value of 4.

21. B

22. B

23. B — Default arguments must be trailing in the parameter list.

24. C

25. C — A function body defined inside a class definition is implicitly inline.

26. B

27. A

28. C

29. B

30. C

31. C — An int literal exactly matches f(int), so it is selected.

32. C — Once a parameter has a default argument, later parameters must also have defaults.

33. B — A recursive function must make progress toward a base case.

34. B

35. B

36. C

37. B

38. C

39. B

40. C

41. B

42. B

43. A

44. B

45. A

46. A

47. C

48. B

49. C

50. A

Unit 2 — Pointers, Reference Variables, Arrays and String Concepts

1. B

2. A

3. A

4. B

5. B

6. A — Pointer arithmetic advances by one element, not one byte.

7. B

8. A

9. C

10. A

11. B — p points to the first array element; *p is 10.

12. B — p+1 points to a[1].

13. C — The pointer moves from a[0] to a[2]; a[2] is 15.

14. B — pp points to p, and *pp dereferences p to x; ++(*pp) increments x.

15. A — The reference r aliases x; changing x changes r.

16. B — The string is assigned the new text before printing.

17. A — s.size() is 4 for "code".

18. B — toupper converts the first character; the example is shown as a direct character conversion.

19. B — The two-dimensional array has a[1][0] equal to 3.

20. A — The object pointer accesses x with ->.

21. B

22. B

23. B

24. B

25. C

26. B

27. B

28. A

29. B

30. B — A pointer-to-data-member uses int A::* pm;

31. B

32. B

33. B

34. B

35. B — a+3 points to the last element; a+4 is one-past and must not be dereferenced.

36. C

37. A

38. B — substr(start, count) starts at index 1 and takes 3 characters.

39. B

40. B — A default memberwise copy duplicates the address, not the resource itself.

41. B

42. A

43. A

44. B

45. B

46. B

47. A

48. A

49. A

50. B

Unit 3 — Data File Operations; Constructors, Destructors and File Handling

1. B

2. B

3. B

4. B — The default behavior for ofstream with ios::out is to truncate unless append is specified.

5. B

6. A

7. C

8. A

9. A

10. B

11. B — The parameterized constructor sets x to the supplied value.

12. A — The default constructor initializes x to 7.

13. B — The copy constructor copies x from a to b.

14. C — The statement prints C, then the destructor prints D when the local object goes out of scope.

15. A — The first constructor prints A and the destructor prints B at scope exit.

16. A — Extraction reads the first whitespace-delimited token.

17. B

18. A — a.x is 1; the copy constructor initializes b.x to a.x+1, so b.x is 2.

19. B

20. A

21. B

22. A

23. B

24. C

25. B

26. B

27. B

28. A

29. A

30. A

31. B

32. A

33. B

34. B

35. B — Test the read operation itself; eof() is usually set only after a read attempt fails.

36. B

37. A

38. C

39. C

40. B

41. B

42. B

43. B

44. C

45. B

46. A

47. A

48. A

49. A

50. B — Automatic objects are destroyed in reverse construction order.

Last-minute revision checklist

Unit I

- class defaults to private; struct defaults to public
- cin uses >>; cout uses <<
- default arguments must be trailing
- reference parameters can modify caller variables
- friend functions are not class members but may access private members
- static data members are shared; define them outside the class (pre-C++17 style shown)
- setw is in <iomanip>; endl inserts newline and flushes
- recursion requires a base case

Unit II

- initialize pointers; use nullptr for an intentional null pointer
- dereference with *; obtain an address with &; object pointer member access uses ->
- pointer arithmetic moves in units of the pointed-to type
- never dereference null, dangling, wild, or one-past pointers
- array indices start at 0; a two-dimensional array uses a[row][column]
- string length()/size(), find(), append(), substr(), erase(), replace()
- a reference must be initialized and cannot be reseated
- void* needs conversion to a suitable object-pointer type before dereferencing

Unit III

- ifstream = input; ofstream = output; fstream = both
- close() closes a stream; test the read operation in while(fin >> x)
- ios::binary selects binary mode; ios::app appends; ios::ate starts at end
- seekg/tellg are for input position; seekp/tellp for output position
- constructors initialize; destructors have ~ClassName() and no return type
- copy constructor commonly takes const ClassName&
- initializer lists initialize members/bases before constructor body
- automatic objects are destructed in reverse construction order

Quick score tracker

Unit	Correct / 50	Wrong / 50	Topics to revise
Unit I			
Unit II			
Unit III			
Total			